



Yong Chang Go

Software Engineer — Semiconductor Manufacturing

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About

Software engineer with 3+ years of experience in semiconductor manufacturing IT, building automation tools, desktop applications, and web systems that support fab operations. Currently a Software Engineer at VSMC focused on engineering data analysis and defect management systems. Previously built MES-integrated tools at UMC handling real-time lot tracking, rack inventory, and legacy system migrations.

Experience

Software Engineer, Data Analysis Platform

Jun 2025 — Present

VisionPower Semiconductor Manufacturing Company (VSMC)

Build and maintain a **KLA Map-to-Map Data Watchdog** in Python — an automated SPC pipeline that monitors active lots in real-time, calculates DC values via formula-driven SQL against KLA inspection data, and submits results to MES with Oracle audit logging and automated retry on failure

Design and develop systems for fab engineering data analysis, defect management, and automated watchdog jobs

Gather requirements directly from engineers and lead end-to-end system delivery — from design through deployment

Evaluate and introduce new tooling to reduce manual data processing and improve operational efficiency

Software Developer, Manufacturing Execution Systems

Feb 2023 — Jun 2025

United Microelectronics Corporation (UMC), Singapore Branch

Developed the **eRack Management System** in C#/.NET (Windows Forms + Oracle SQL + multi-threading), enabling check-in/check-out tracking of racks and carriers with real-time MES integration for inventory accuracy

Built an **Automated WAT Rework and Backup System** in C# integrated with Aegis and ADO.NET, automating lot verification, hold code management, and parameter modifications to reduce manual intervention — with enhanced error handling via automated email notifications, error holding lot, and hold comments

Migrated the **Reticle Management System** from VB to C#/ASP.NET Web Forms, implementing AJAX partial updates and session management while maintaining full feature parity

Maintained and enhanced existing MES applications, collaborating with production staff to diagnose issues and deploy fixes with minimal downtime

IT Support Intern

Mar — Jul 2021

Vis-Dynamics Research Sdn Bhd, Malacca

Vendor IT support across networking and server infrastructure; implemented security hardening measures.

Technical Skills

LANGUAGES

C# Python SQL

HTML/CSS/JavaScript

VB.NET SAS

FRAMEWORKS

.NET Framework

ASP.NET Web Forms

Windows Forms

ADO.NET

TOOLS & PLATFORMS

Git Gitea

Azure DevOps

DBeaver

Oracle SQL Developer

DOMAIN EXPERTISE

Semiconductor
Manufacturing

MES Integration

SPC Data Pipelines

Process Automation

Education

Bachelor of Information Technology (Honours) — Artificial Intelligence

Jun 2018 — Sep 2022

Multimedia University (Malacca Campus), Malacca

CGPA: 3.36